

Discussion 2- Solutions

1. List each of the 16 sequences in the sample space S.

<i>HHHH</i>	<i>HTHH</i>	<i>THHH</i>	<i>HHHT</i>
<i>HHTH</i>	<i>HTHT</i>	<i>THHT</i>	<i>HHTT</i>
<i>TTHH</i>	<i>HTTH</i>	<i>THTH</i>	<i>TTHT</i>
<i>TTTH</i>	<i>HTTT</i>	<i>THTT</i>	<i>TTTT</i>

We have the following events

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>HHHH</i>	<i>HHTT</i>	<i>HHHH</i>	<i>HTTT</i>
<i>HHTH</i>	<i>HTHT</i>	<i>THHH</i>	<i>THTT</i>
<i>HHHT</i>	<i>HTTH</i>	<i>HTHH</i>	<i>TTHT</i>
<i>HTHH</i>	<i>TTHH</i>	<i>TTHH</i>	<i>TTTH</i>
<i>THHH</i>	<i>THHT</i>	<i>HHHT</i>	
	<i>THTH</i>	<i>THHT</i>	
	<i>HTTT</i>	<i>HTHT</i>	
	<i>THTT</i>	<i>TTHT</i>	
	<i>TTHT</i>		
	<i>TTTH</i>		
	<i>TTTT</i>		

Hence, $P(A) = \frac{5}{16}$, $P(B) = \frac{11}{16}$, $P(C) = \frac{8}{16}$, $P(D) = \frac{4}{16}$.

$A \cap C = \{HHHH, HHHT, HTHH, THHH\}$, $P(A \cap C) = \frac{4}{16}$,

$B \cap D = D$, $P(B \cap D) = P(D) = \frac{4}{16}$.

2. (a)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.7 = 0.4 + 0.5 - P(A \cap B)$$

$$P(A \cap B) = 0.2.$$

(b) $A' \cap B' = (A \cup B)'$,

$$P((A \cup B)') = 1 - P(A \cup B) = 0.3$$

$$(c) A' \cup B' = (A \cap B)'$$

$$P((A \cap B)') = 1 - P(A \cap B) = 0.8$$

$$3. (i) E = \text{Sum of the two die are } 3 = \{(1,2), (2,1)\}, P(E) = \frac{2}{36}.$$

$$(ii) A = 6 \leq \text{Sum of the two die} \leq 8,$$

$$A = \{(3,3), (1,5), (5,1), (2,4), (4,2), (2,5), (5,2), (1,6), (6,1), (3,4), (4,3), (4,4), (2,6), (6,2), (3,5), (5,3)\},$$

$$P(A) = \frac{16}{36} = \frac{4}{9}$$

$$4. P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, $P(A \cap B) = 0$. No, A and B cannot be mutually exclusive since $P(A) + P(B) = 1.1 > 1$.

$$P(A \cup B) \leq 1.$$

$$\text{Hence, } P(A) + P(B) - P(A \cap B) \leq 1.$$

$$\text{Hence, } P(A) + P(B) - 1 \leq P(A \cap B).$$

$$\text{Hence, } 0.1 \leq P(A \cap B). \text{ Hence minimum of } P(A \cap B) = 0.1.$$

Generally let's say we have two sets E, F. When we calculate the $E \cap F$, the largest answer happens when one of them is the subset of the other one. For example if $E \subseteq F$, then $E \cap F = E$. $A \cap B$ in the largest possible scenario is B, because B has a smaller probability. Hence, $P(A \cap B) = 0.5$.

5.

$$(a) P(A \cup B \cup C) = 0.75.$$

$$(b) P((A \cap B) \cup C) = 0.67.$$

$$(c) P(A \cap (B \cup C)) = 0.37.$$